SWEN30006
Software Modelling and Design

GRASP:
OO Design with Responsibilities

Textbook: Larman Chapter 17

“Understanding responsibilities is key to good object-oriented design"

—Martin Fowler

Learning Objectives

On completion of this topic you should be able to:

- Understand the concepts of responsibility and responsibility-driven design
- Understand the advantages of using patterns
- Apply five of the GRASP principles or patterns for Object-Oriented Design.

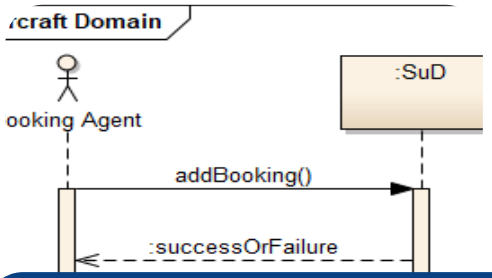
Revisited: Object-Oriented Analysis, Design, and Development

Problem domain

- Concepts
- Relationships

Use Cases

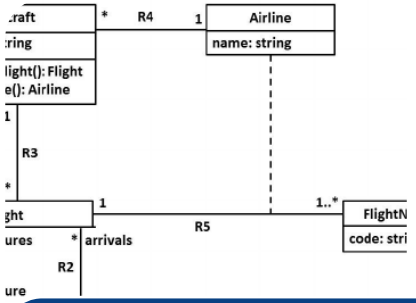
Text stories
Week 1



OO Domain Models

- Static: Domain Class Diagram
- Dynamic: System Sequence Diagram

Conceptual Classes
Week 2



OO Design Models

- Static: Design Class Diagram
- Dynamic: Design Sequence Diagram

Software Classes
Week 3

```
d do_important_stuf
teZInputs();
t = zalg.apply_ZAlg(
stZOutput(zoutput))
eZOutput();
{
```

OO Implementation

Classes in Programming
Week 3

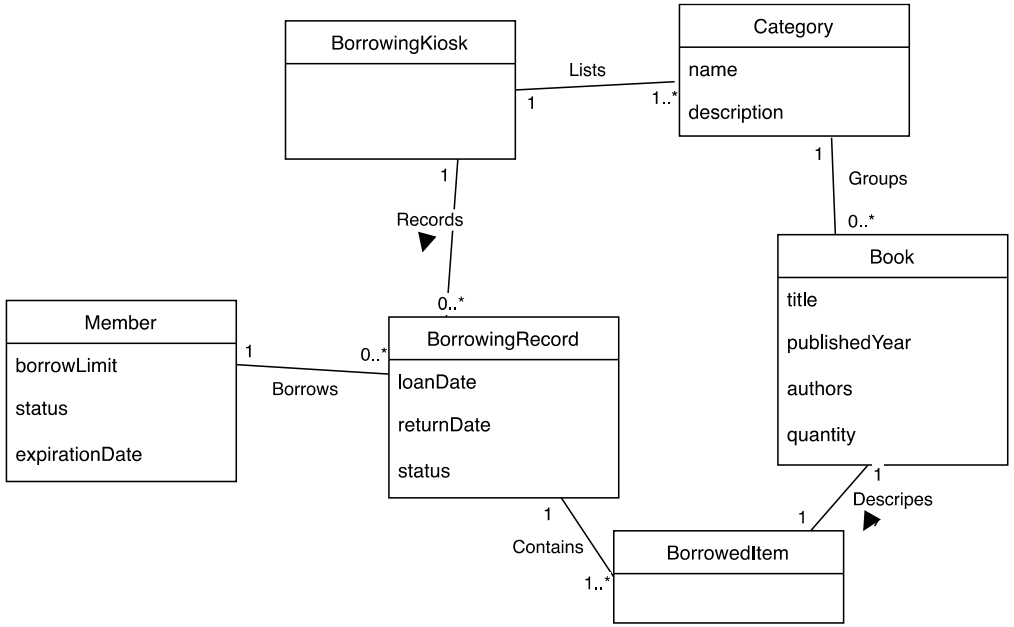
Revisited: Responsibility-Driven Design (RDD)

- RDD focuses on assigning *responsibility* to software objects
  - Responsibility: The obligations or behaviours of an object in terms of its role
- Two types of responsibility:

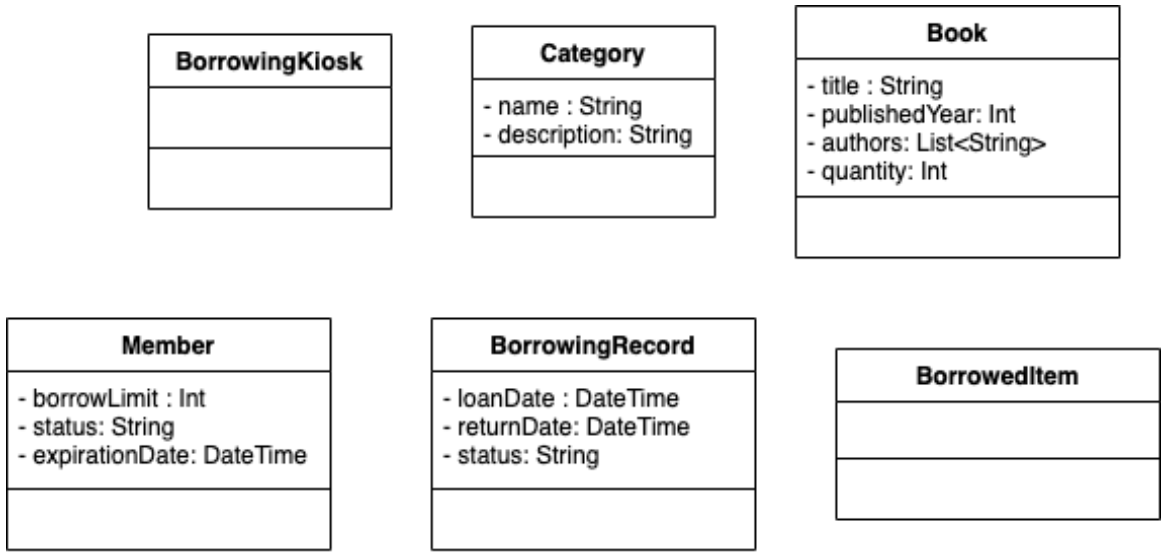
- **Knowing** responsibilities include:
  - knowing about private encapsulated data
  - knowing about related objects
  - knowing about things it can derive or calculate

- **Doing** responsibilities include:
  - doing something itself, such as creating an object or doing a calculation
  - initiating action in other objects
  - controlling and coordinating activities in other objects

Example: Designing the design model based on the domain model



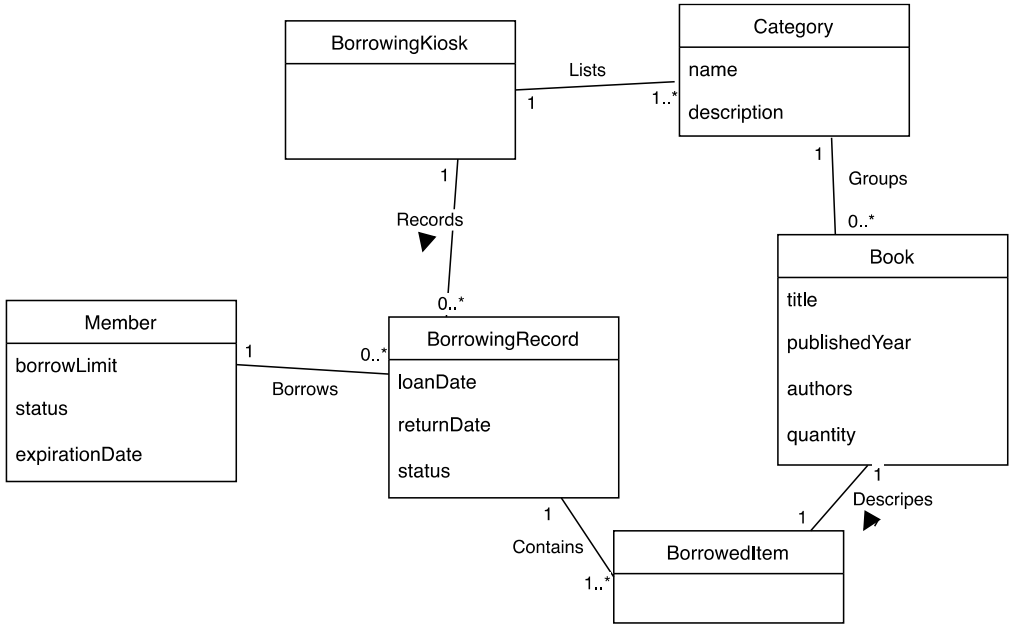
A **domain model** of
Library Borrowing System



An *incomplete* **design model** of
Library Borrowing System

Q1: Who should be responsible for creating “BorrowedItem” object?

Example: Designing the design model based on the domain model



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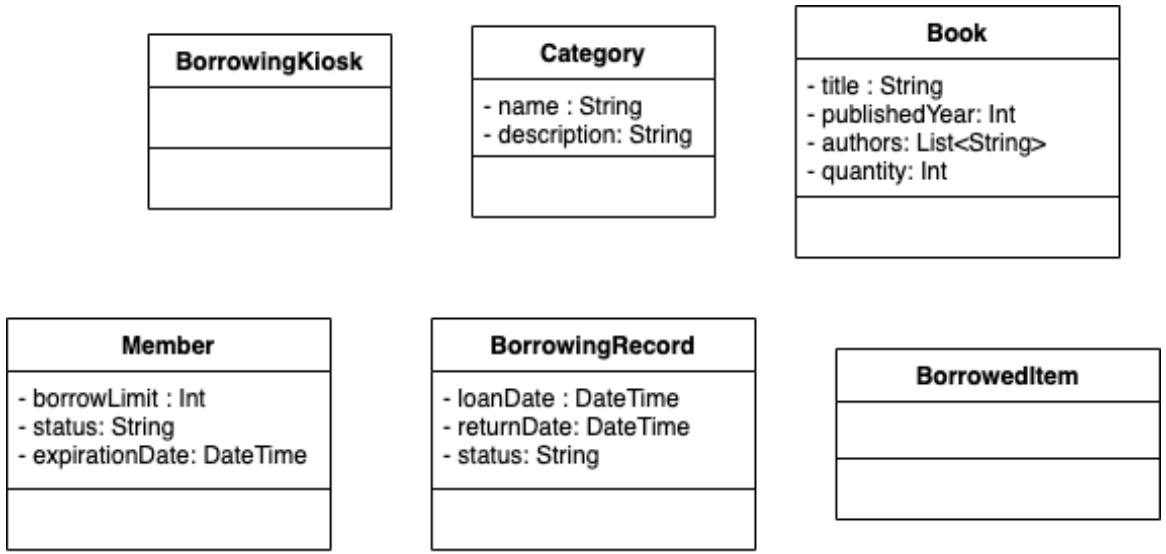

An *incomplete* **design model** of Library Borrowing System

Q2: Who should be responsible for checking if a member reaches the borrowing limit?

GRASP: General Responsibility Assignment Software Patterns/Principles

- **GRASP – Definition:** A set of patterns (or principles) of assigning responsibilities into an object.
- GRASP aids in applying design reasoning in a methodical, rational, and explainable way
  - Given a specific category of problem, GRASP guides the assignment of responsibilities to objects
- Useful to know and support Responsibility-Driven Design (RDD)
- **Pattern – Definition:**
  - A *named* and *well-known* problem/solution pair that can be applied in new contexts
  - A recurring successful application of expertise in a particular domain

Some patterns in real life

- Q: What should a bathroom have?
- A: It should have a wash basin, a bathtub, a toilet, etc.

- Q: Where can I cross the road?
- A: Looking for pedestrian crossing sign

- Q: How should I use the area near the entrance of a building (like Stop1)?
- A: Make it a reception area

Advantages of Patterns

"To become a master, one must study the successful results of other masters!"

- *Capture expertise* and make it accessible to non-experts in a standard form
- Facilitate successful applications of expertise is *easily re-used*
- *Improve understandability*
- Facilitate communication among practitioners by *providing a common language*
- A new solution can be generated based on a modified version of existing patterns

GRASP: Learning Aid for RDD

- This subject will cover the 9 GRASP patterns/principles
  - Creator
  - Information Expert
  - Low Coupling
  - High Cohesion
  - Controller
  - Polymorphism
  - Indirection
  - Pure Fabrication
  - Protected Variations

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*Controller will be covered in Workshop supplementary material

GRASP: Creator

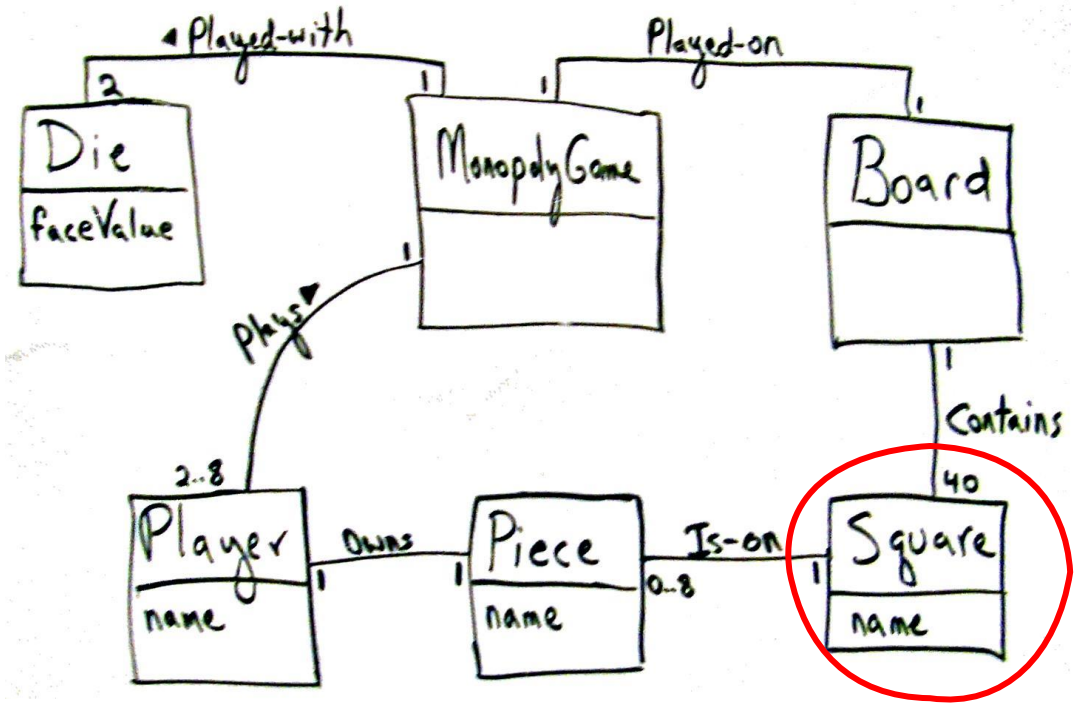
Creator

- **Problem:** Who should be responsible for creating a new instance of class *A*? *(Doing Responsibility)*
  - Creation of objects is one of the most common OO activities
  - Useful for lower maintenance and higher opportunities for reuse (Low Coupling)
- **Solution:** Assign class *B* responsibility to create instances of class *A* *if* one of these is true (the more the better):
  - *B* "contains" or compositely aggregates *A*
  - *B* records *A*.
  - *B* closely uses *A*.
  - *B* has the initializing data for *A*.

*To achieve **low representational gap**, we use the domain model to inspire the design classes*

*If more than one class could be the creator, prefer the class which contains/aggregates *A*

Example: Monopoly



Monopoly Game Simulation Domain Model

Problem: Who should be responsible for creating 'Square' object?

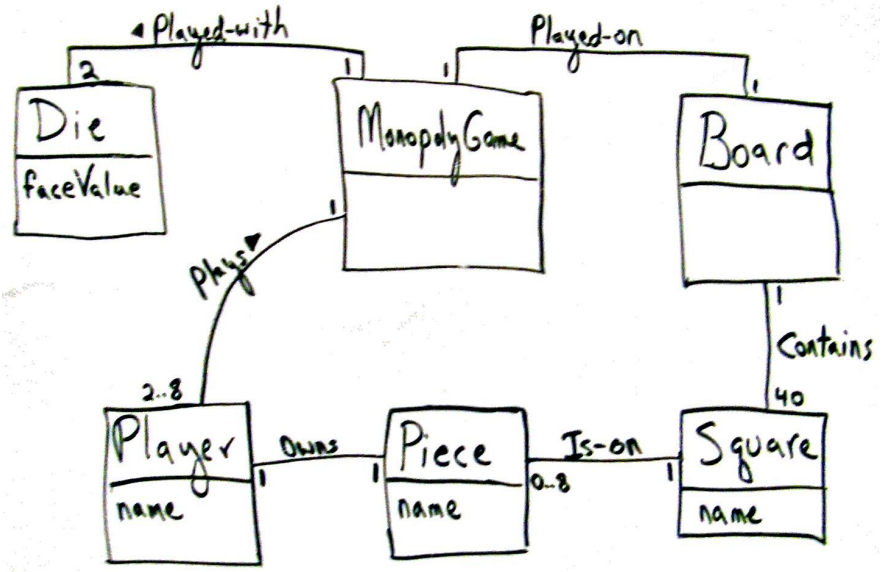
Solution:

- *Who* "contains" or compositely aggregates *Square objects* → ***Board***
- *Who* records *Square objects* → ***N/A***
- *Who* closely uses *Square objects* → ***Board & Piece***
- *Who* has the initializing data for *Square objects* → ***N/A***

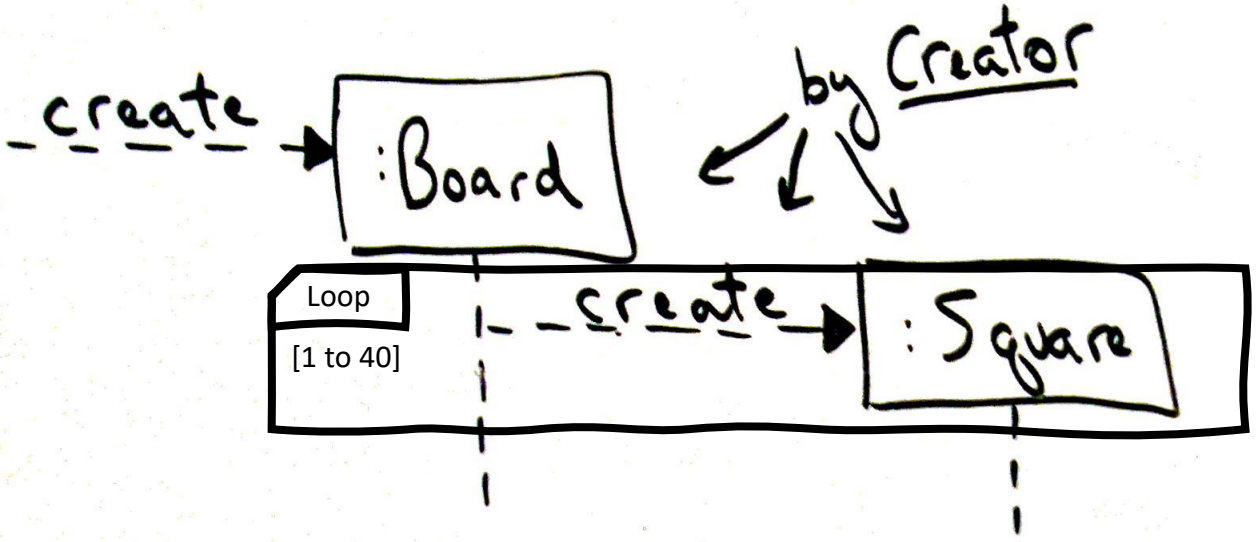
***Board* class should be responsible for creating *Square* object**

Example: Monopoly

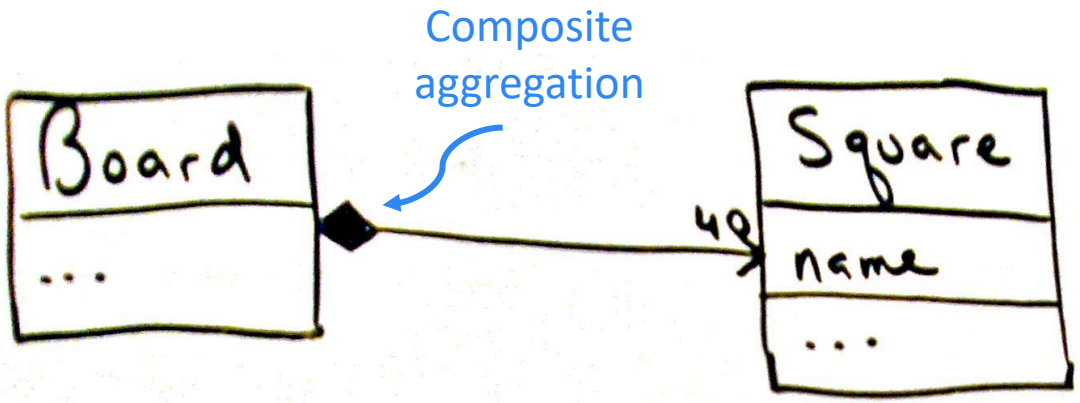
Board **class should be responsible for creating *Square* object**



Monopoly Game Simulation Domain Model



Design Sequence Diagram

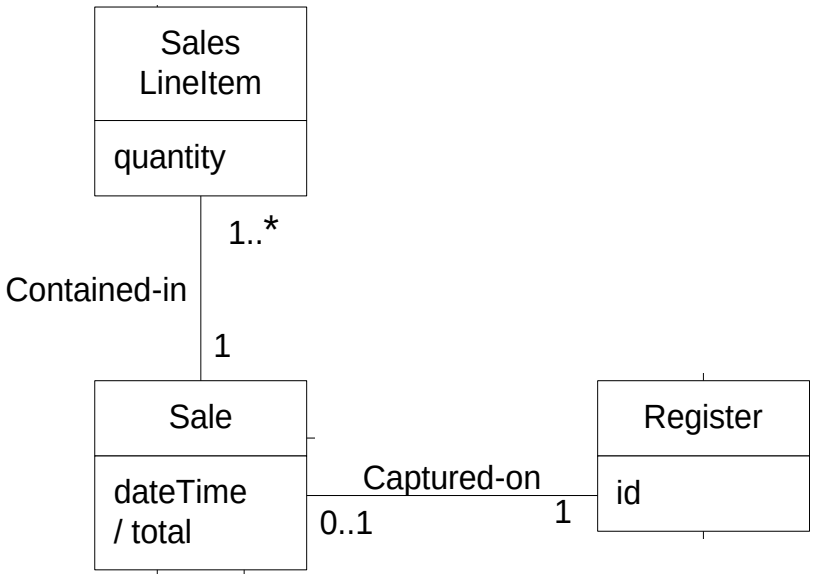


Design Class Diagram

More Examples

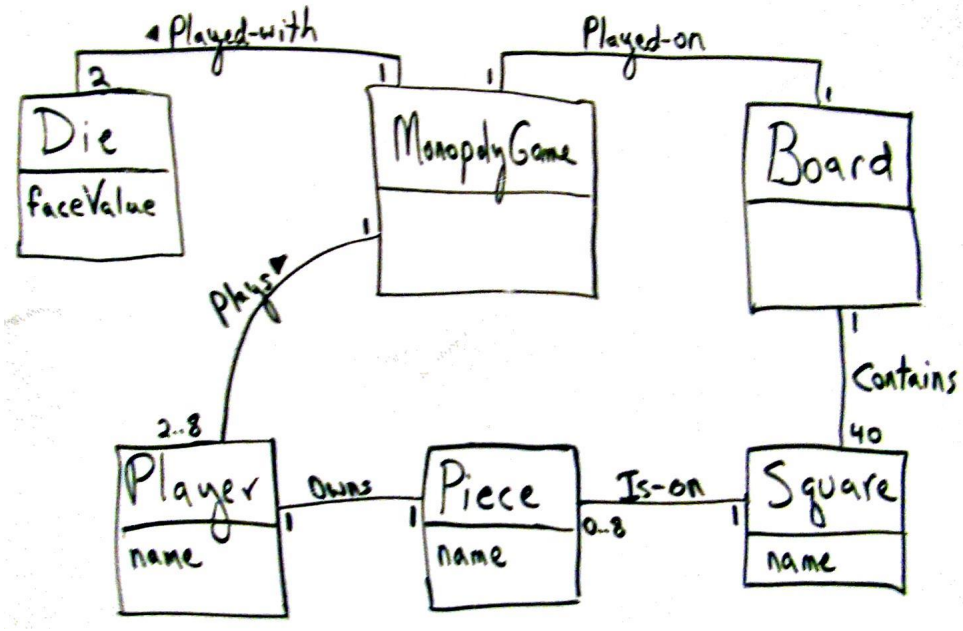
Who is responsible for creating "SaleLineItem" object?

→ Since a *Sale* class contains/aggregates many *SaleLineItem* objects, Sale should create *SaleLineItem* objects



Who is responsible for creating "Player" object?

→ Since a MonopolyGame class has the initializing data for "Player" object (i.e. the Player's name received from the user input), MonopolyGame should create 'Player' object



Creator: Contraindications

- If creation of objects has significant complexity, e.g.,
  - Using recycled instances for performance
  - Conditionally creating an instance from one of a family of similar classes based upon some external property value or other context information.
- It is advisable to delegate creation to a helper class called a *Concrete Factory* or an *Abstract Factory* rather than use the class suggested by Creator
- Concrete Factory will be taught in GoF Design Patterns lectures (Weeks 7-9)

GRASP: Information Expert (or Expert)

Information Expert (or Expert)

- **Problem:** What is a general principle of assigning responsibilities to objects?
  - A design model may have 100s (or 1,000s) software classes with 100s (or 1,000s) of responsibilities to be fulfilled
  - Given object *X,* which responsibilities (e.g., doing something or knowing something) can be assigned to *X*?
- **Solution:** Assign the responsibility to object *X* if *X* has the information to fulfill that responsibility
  - Information Expert is useful for understandability, maintainability, and extendibility

Where to look?:

1. *If there is relevant class in the Design model, look there first*
2. *Otherwise, look in the Domain model to inspire creation of design classes*

Example: Monopoly

Suppose there are objects that need to be able to reference a particular *Square*, given its name.

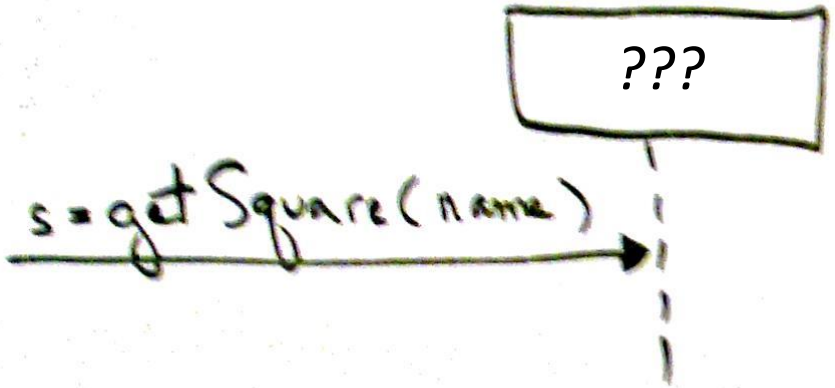
Who should be responsible for identifying a *Square*, given a key?

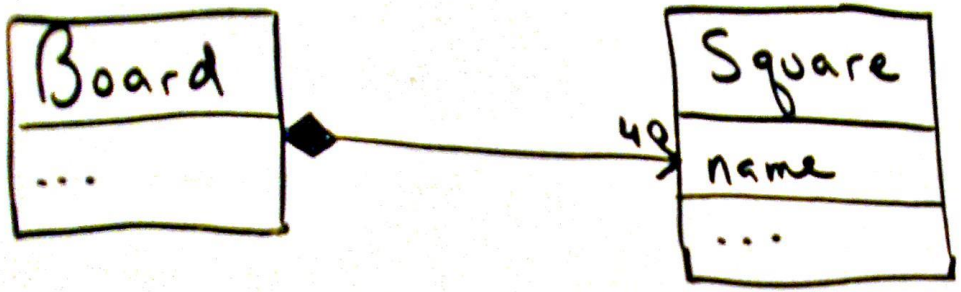
<u>Approach:</u>

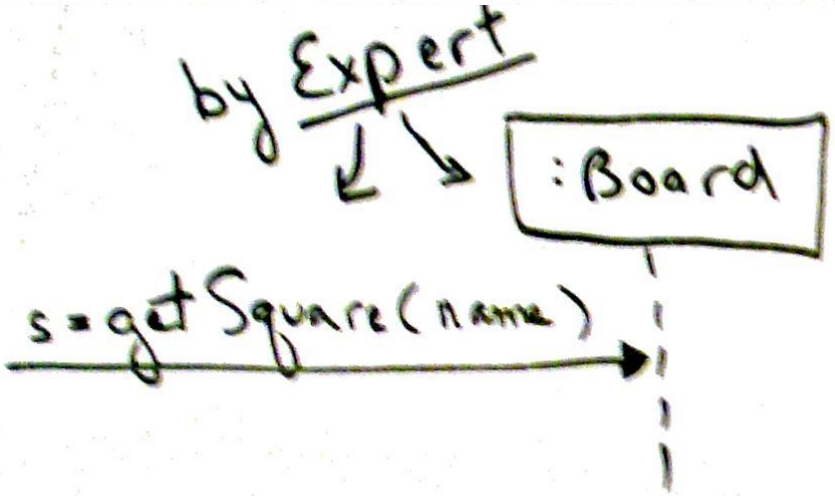
→ Who has information about a Square? Who can access Square objects?

→ The software *Board* class aggregates all the *Square* objects (based on the previous example).

<u>Solution:</u> Board has information about 'Square' (i.e., is an information expert). Thus, Board should be responsible for identifying a Square, given a key.







Another Example: POS

Who should be responsible for calculating *Subtotal*?

Approach:

→ What information do we need for 'Subtotal'?

→ Subtotal = quantity * price

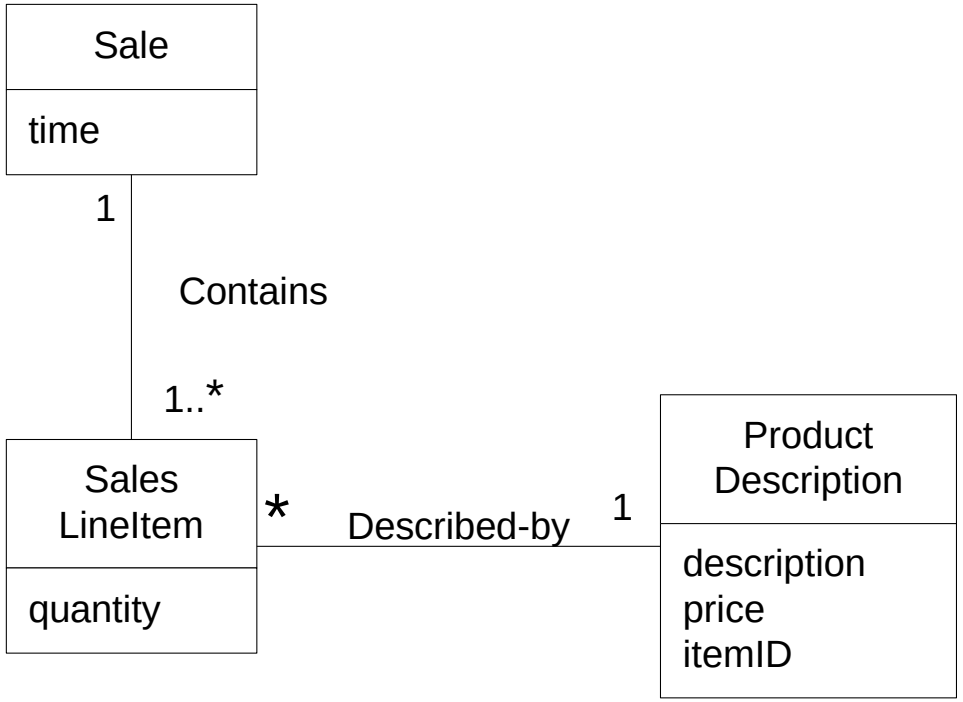
→ Who has (or can access) information about quantity and price?

→ SaleLineItem knows quantity, ProductDescription knows price

Also an Information Expert

→ SaleLineItem associates with ProductDescription, then SaleLineItem can also access price

Solution: By Expert, SaleLineItem should be responsible for calculating Subtotal



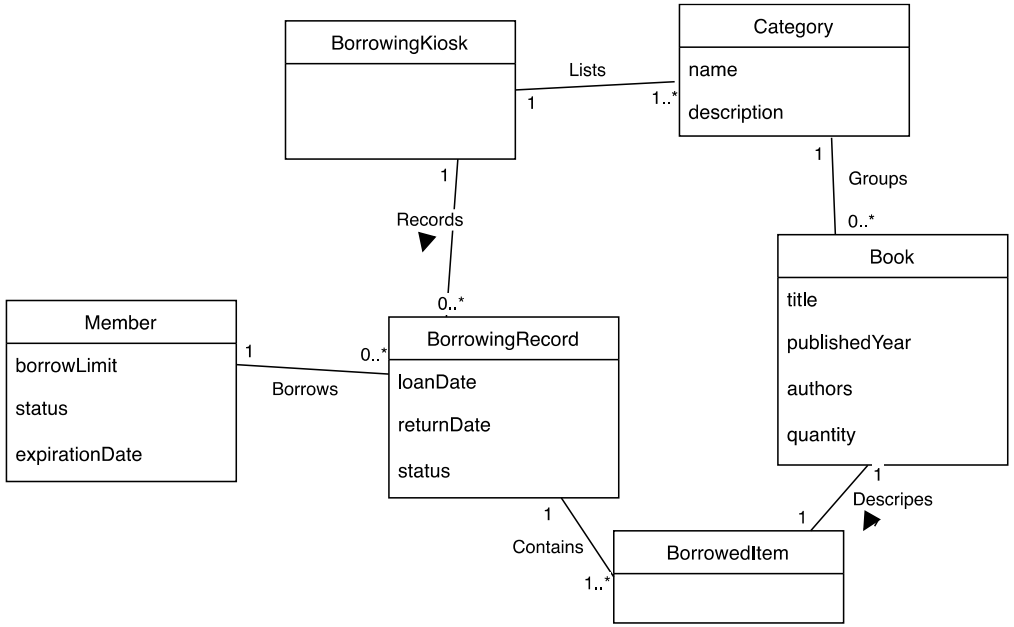
Information Expert: Contraindications

In some situations, a solution suggested by Expert is undesirable, usually because of problems with the resulting dependencies.

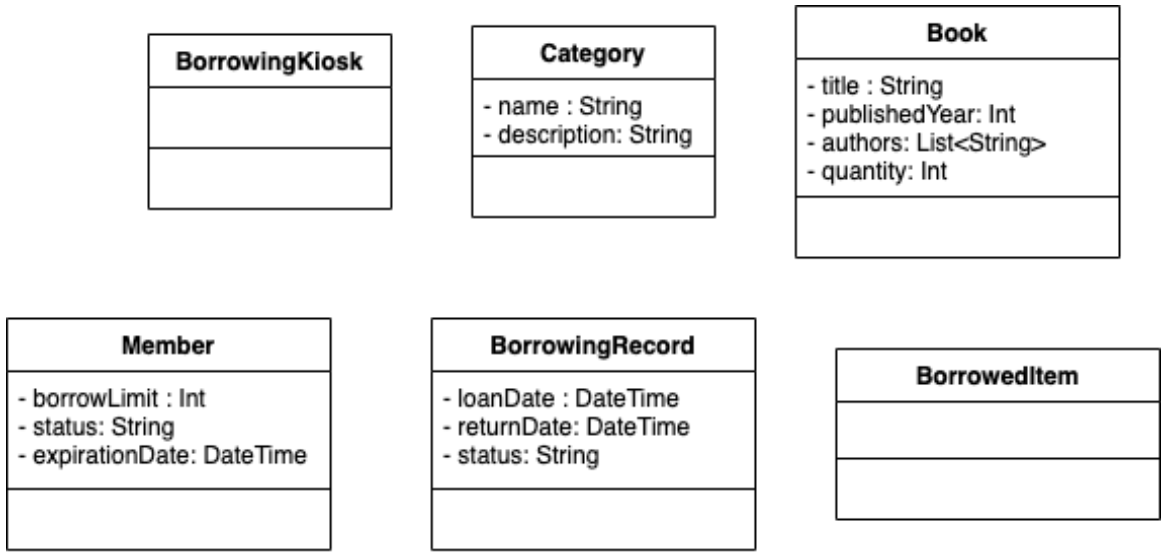
Example: Who should be responsible for storing a Sale transaction in the database (DB)?

- By Expert, Sale should be responsible
- But, DB handling (e.g. SQL) is not related to the logic of the application
- Putting DB logic in a class inspired by the domain increases the representational gap
- Putting DB logic in every class that needs to be saved in the DB would create lots of classes with dependencies on the DB technology -> poor maintainability
- DB logic should be done by other helper classes: DB expert classes

Example: Designing the design model based on the domain model



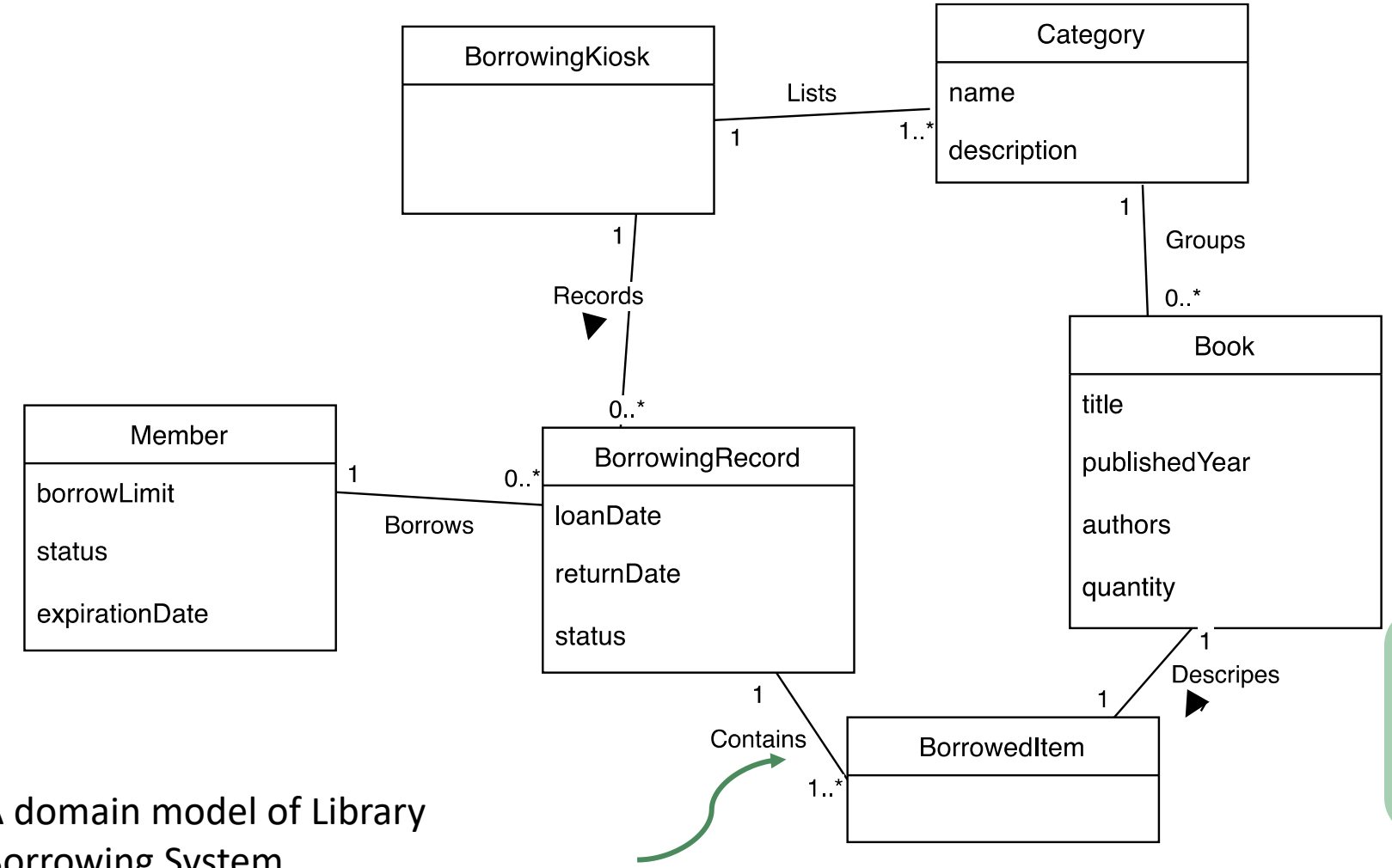
A **domain model** of
Library Borrowing System



An *incomplete* **design model** of
Library Borrowing System

Who should be responsible for creating
“BorrowedItem” object?

Exercise: Assigning responsibilities to software objects of the design model based on the domain model



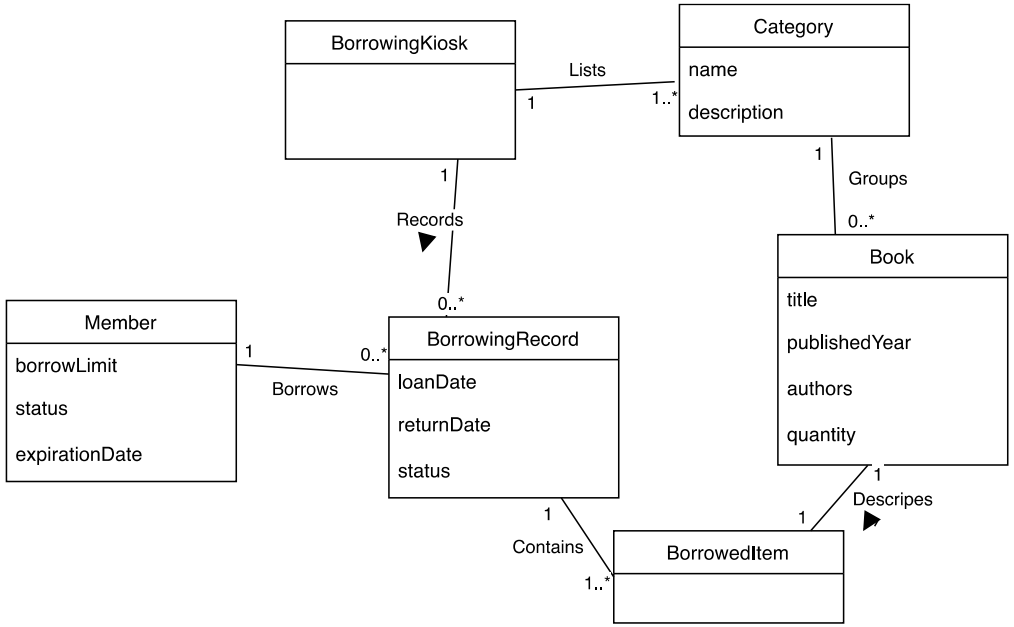
A domain model of Library Borrowing System

Inspired by the domain model, BorrowingRecord contains/compositely aggregate BorrowedItem

Who should be responsible for creating “BorrowedItem” object?

Suggested by Creator, BorrowingRecord should be responsible for creating BorrowedItem object.

Example: Designing the design model based on the domain model



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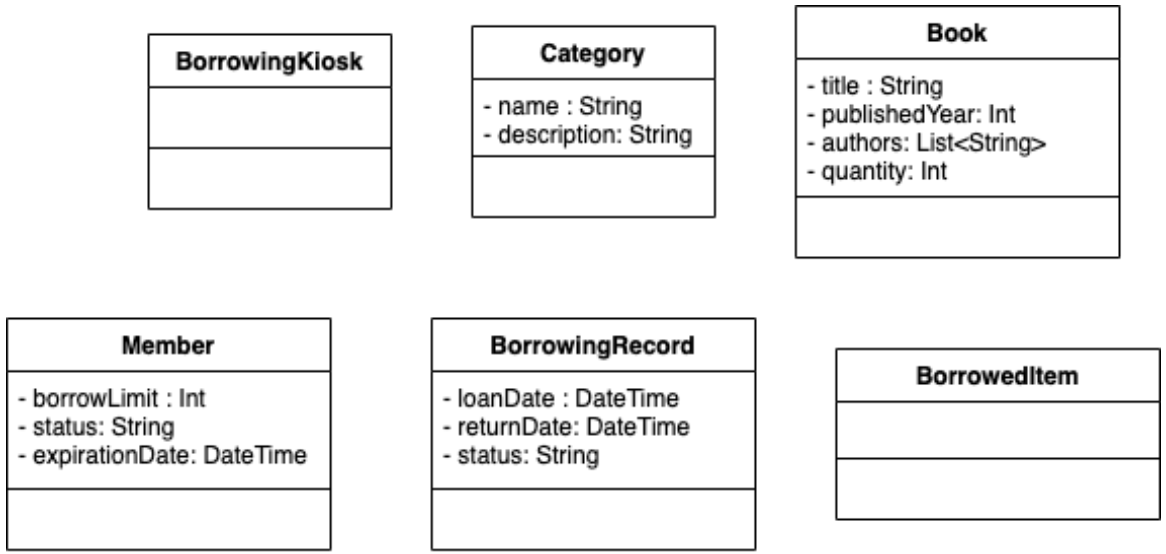

An *incomplete* **design model** of
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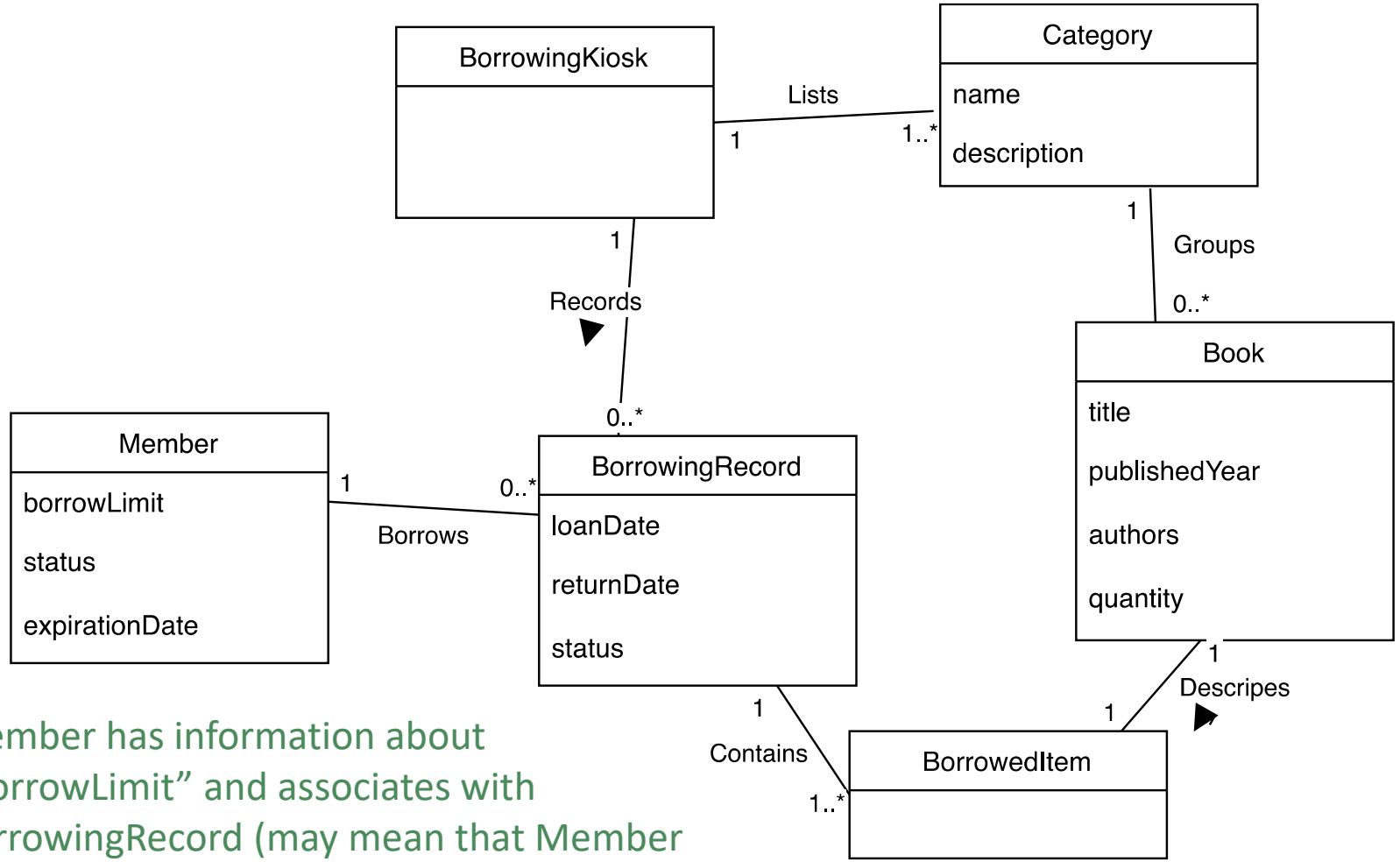
Who should be responsible for checking if a member reaches the borrowing limit?

Exercise: Assigning responsibilities to software objects of the design model based on the domain model



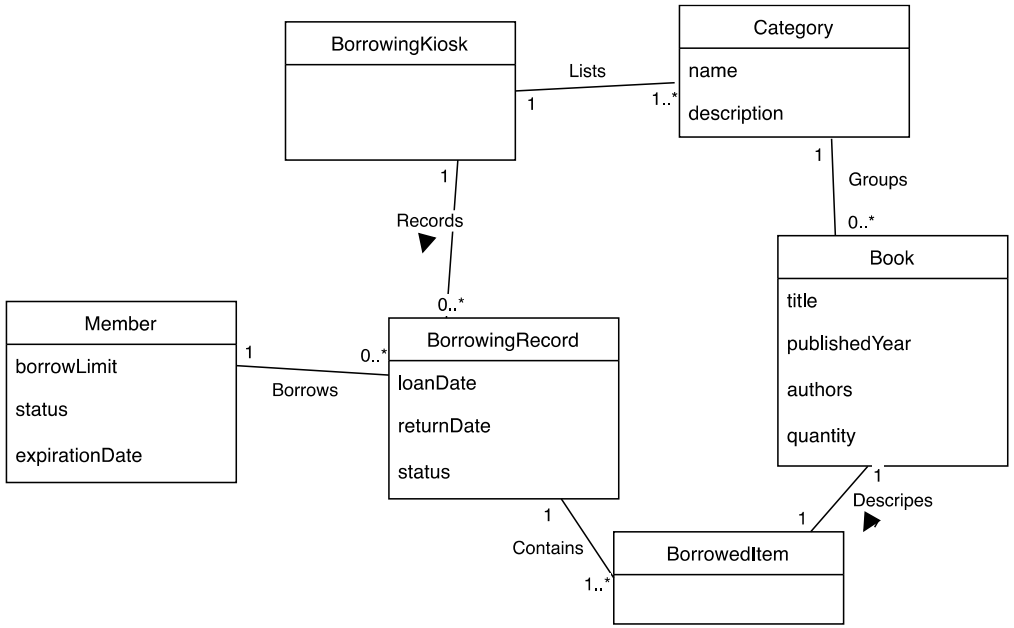
Member has information about "borrowLimit" and associates with BorrowingRecord (may mean that Member can also own BorrowingRecord)

Who should be responsible for checking if a member reaches the borrowing limit?

By Expert, Member should be responsible for checking if a member reaches the borrowing limit

NOTE: Other designs might be applicable. Use other GRASP principles to evaluate the alternatives

Example: Designing the design model based on the domain model



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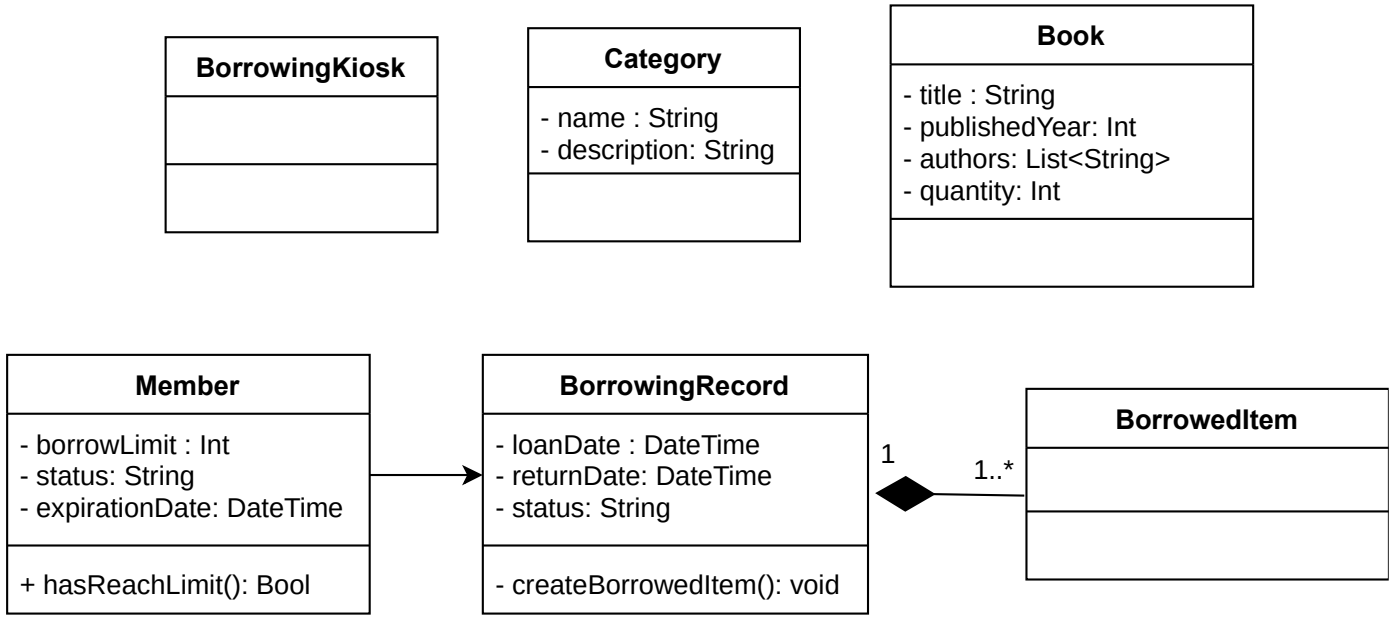


An *incomplete* **design model** of Library Borrowing System

Why can't the BorrowingKiosk class have responsibility of "hasReachedLimit" instead of the Member class?

GRASP:
Low Coupling

Low Coupling

- **Problem:** How to support low dependency, low change impact, and increased reuse?
  - Coupling = how strongly one element is connected to (has knowledge of, or relies on) other elements
  - Low Coupling = Each element is not dependent on too many other elements
  - High Coupling = Hard to understand in isolation, hard to reuse, one change impacts many classes (high impact change)

- **Solution:** Assign responsibilities to avoid unnecessary coupling and to keep coupling low
  - *Use this principle to evaluate alternatives*
  - Low Coupling minimizes dependencies, making the system more maintainable, and the code more reusable

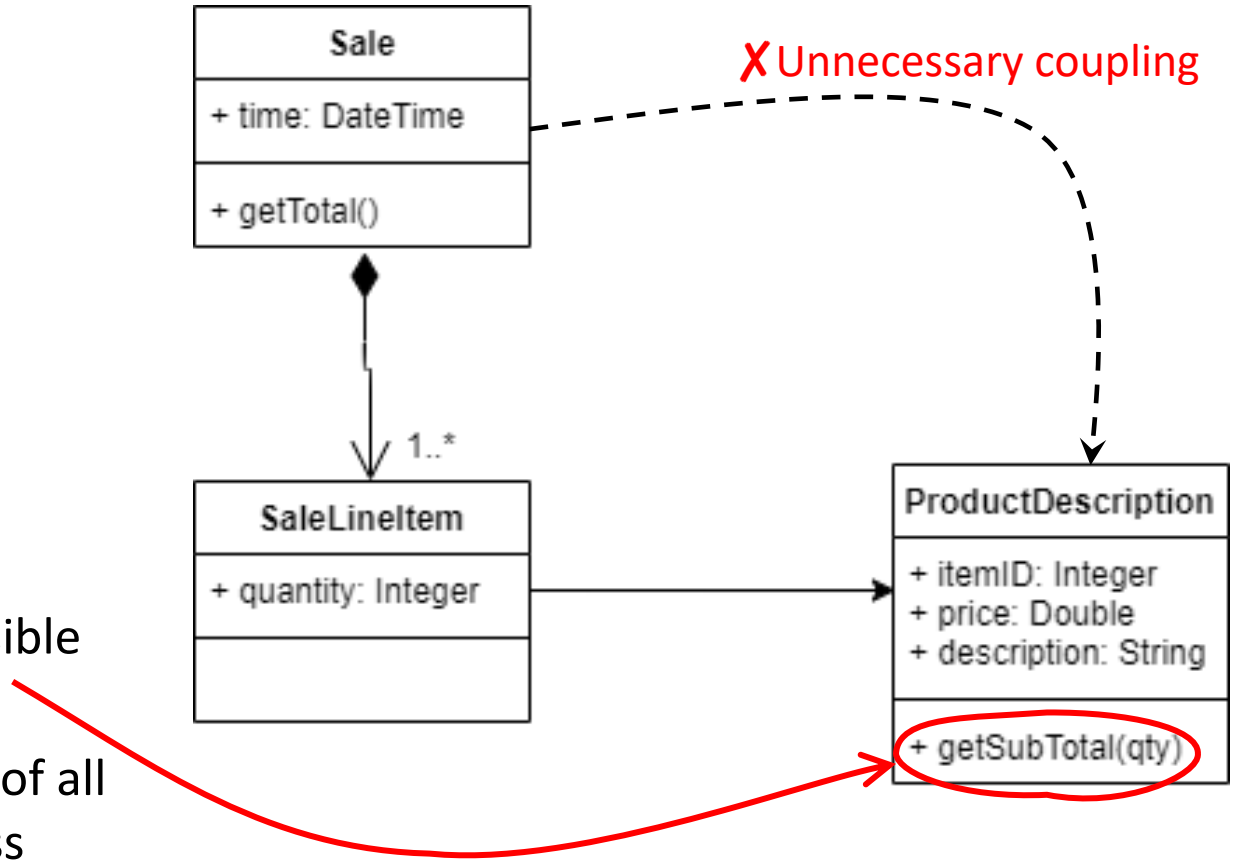
Example: POS

Who should be responsible for calculating *SubTotal*?

- SaleLineItem knows quantity, ProductDescription knows price

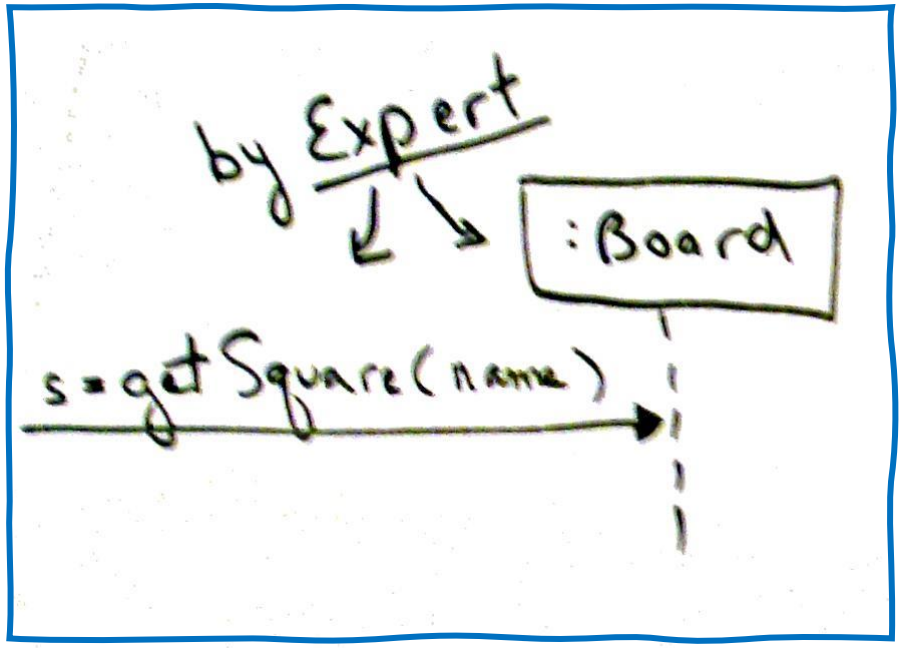
An alternative solution:

- By Expert, ProductDescription can also be responsible to calculate SubTotal
  - → Yes. But when Sale object will sum the SubTotal of all SaleLineItem objects, Sale object needs to access ProductDescription
  - → The Low Coupling principle prefers SaleLineItem to calculate SubTotal.

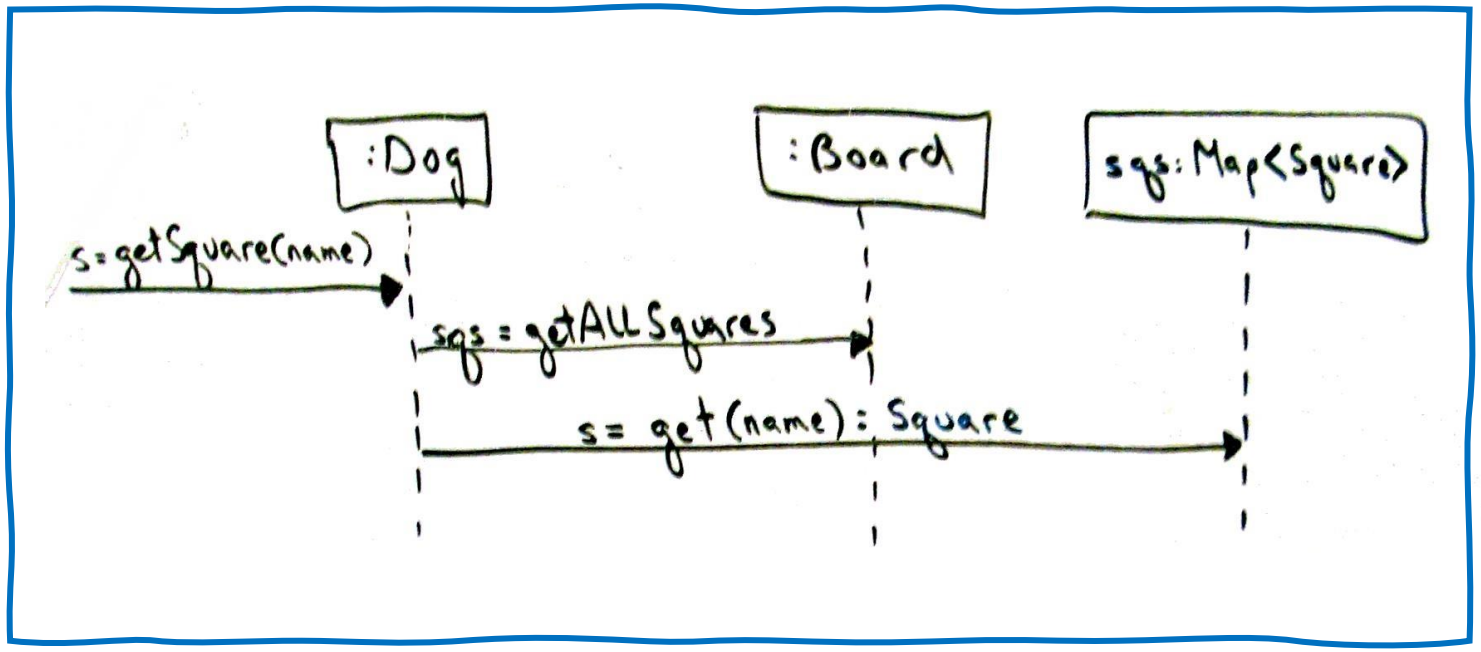


Another Example: Monopoly

Why can't we have other class (like Dog; an arbitrary class) to get a Square object instead of Board?



Design A



Design B

Another Example: Monopoly

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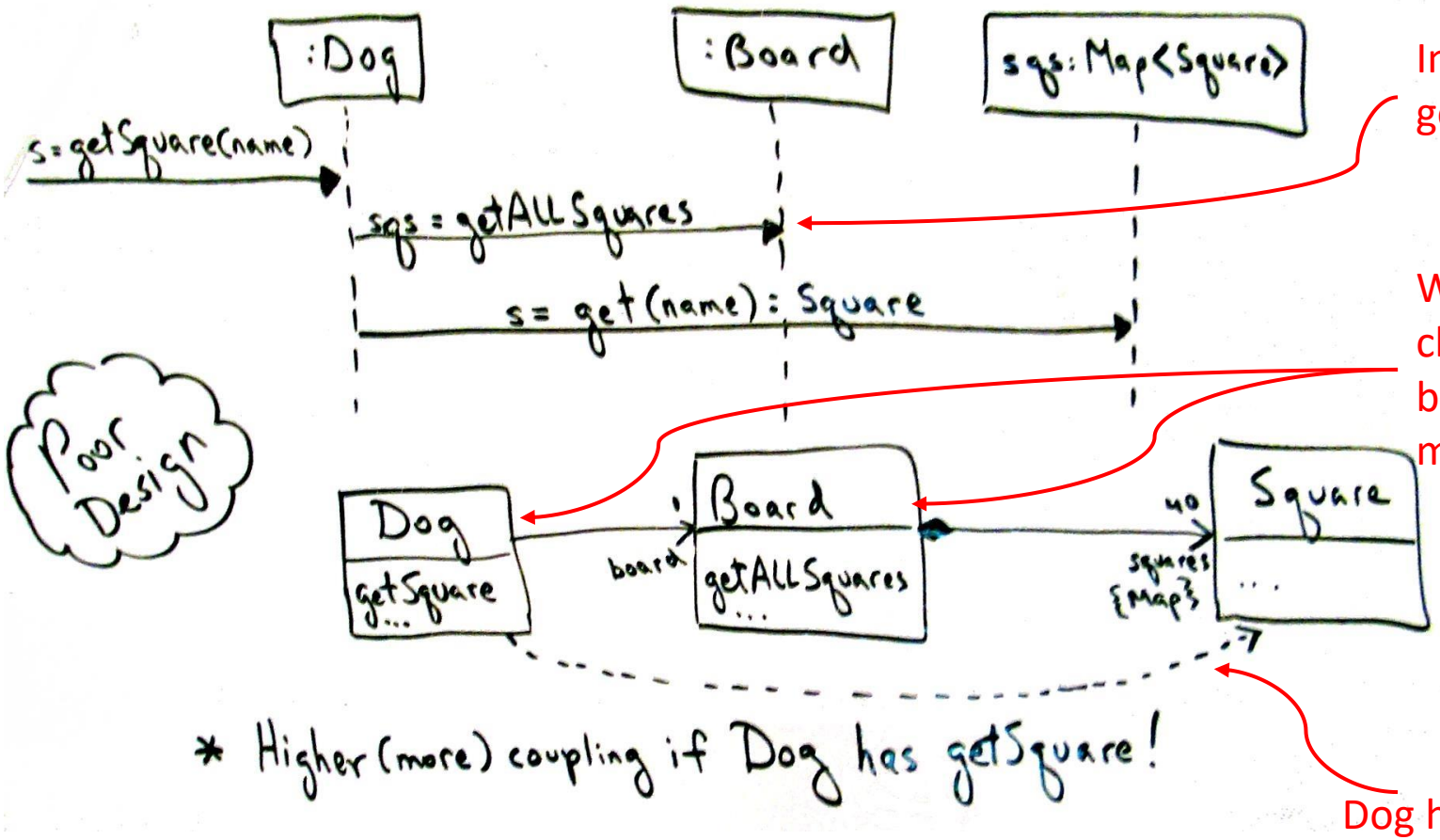


Increase an additional step to get Square → Not efficient

When an interface in Square is changed, forcing a change in both Dog and Board → Hard to maintain

Dog has to connected with Square to obtain knowledge

Low Coupling: Discussion

- Low Coupling is an ***evaluative principle*** that should be kept in mind during all design decisions
- Low Coupling supports the design of classes that are more independent, reducing the impact of change
- Low Coupling help you avoid increasing unnecessary dependencies/coupling that could lead to a negative result

Contraindications

- High coupling with stable and well understood elements (e.g., Standard Java Libraries) is not usually a problem: few issues with propagating changes, understandability, and maintainability.

High Cohesion

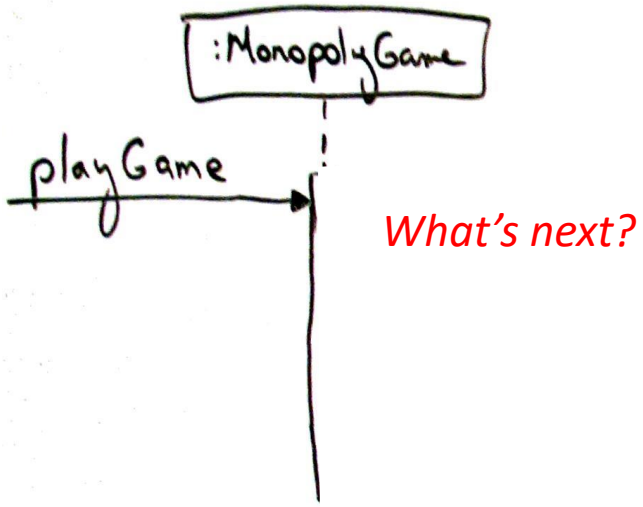
High Cohesion

- **Problem:** How to keep objects focused, understandable, and manageable, and as a side effect, support Low Coupling?
  - (Functional) Cohesion = How strongly related and focused the responsibilities of an element are
  - Low cohesion = a class has many unrelated things or does too much work, resulting code to be hard to comprehend, reuse, maintain

- **Solution:** Assign responsibilities to avoid unnecessarily reducing cohesion and to keep cohesion high
  - *Use this principle to evaluate alternatives*
  - High cohesion will ease of comprehension of the design, simplify maintainability and enhancement

Example: Monopoly

When there is an input of playGame to MonopolyGame object, what MonopolyGame should do next?



What's next?

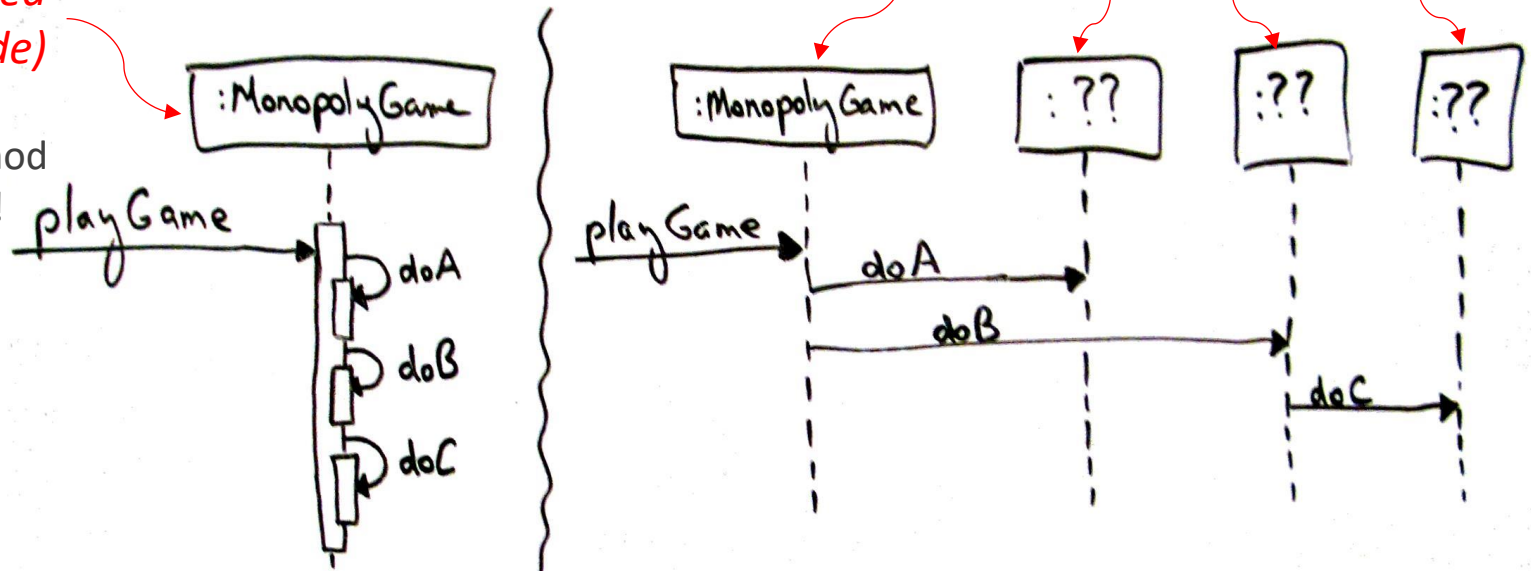
Example: Monopoly

When there is an input of playGame to MonopolyGame object, what MonopolyGame should do next?
-- Should MonopolyGame object be responsible for doing all the tasks, or delegate some tasks to other object?

The class can be bloated (too much code)

Imagine a class has 100 methods, and each method has 200 lines = 20K lines!

The code is more focused. Easy to understand, and maintainable



Do all the tasks?

Delegate some tasks to others?

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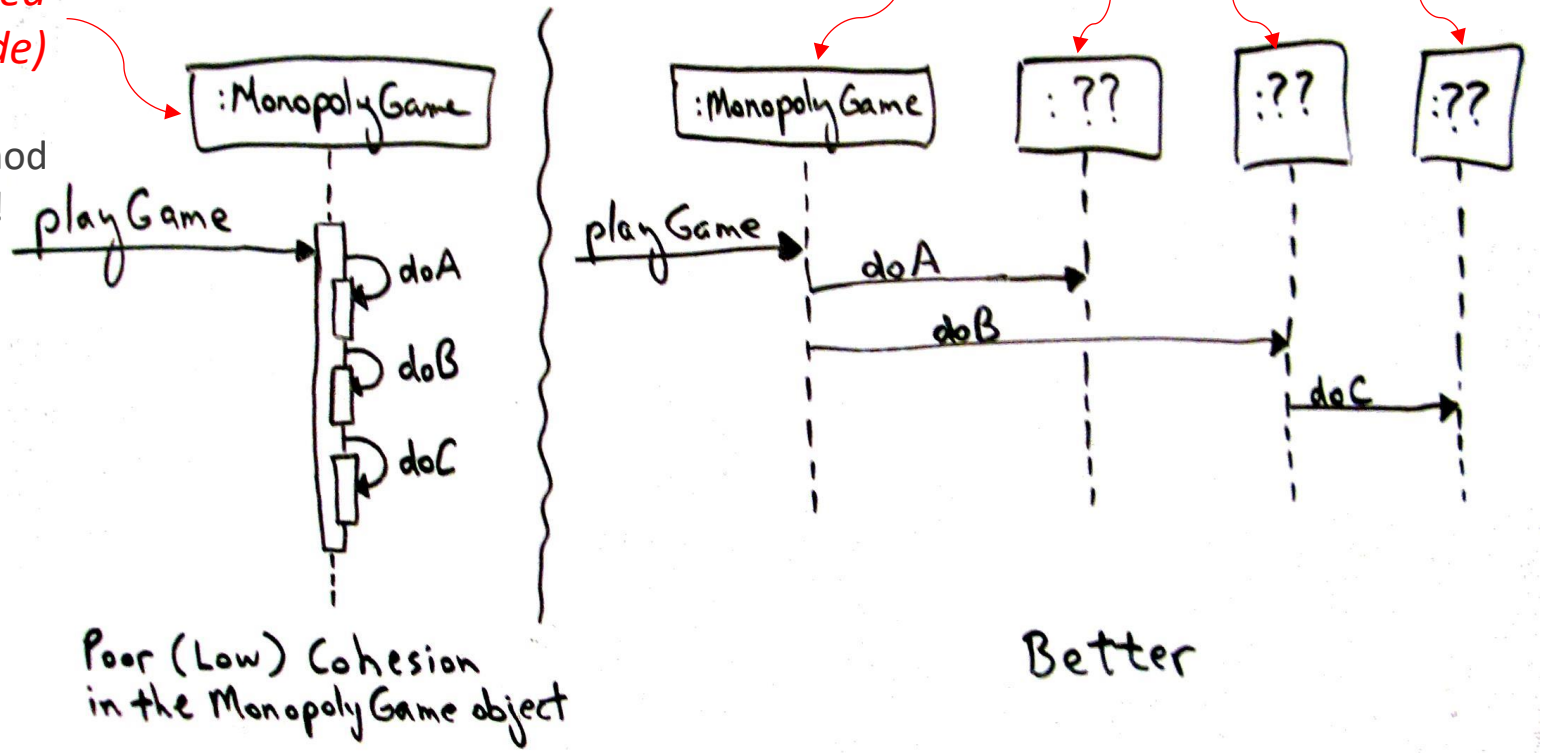
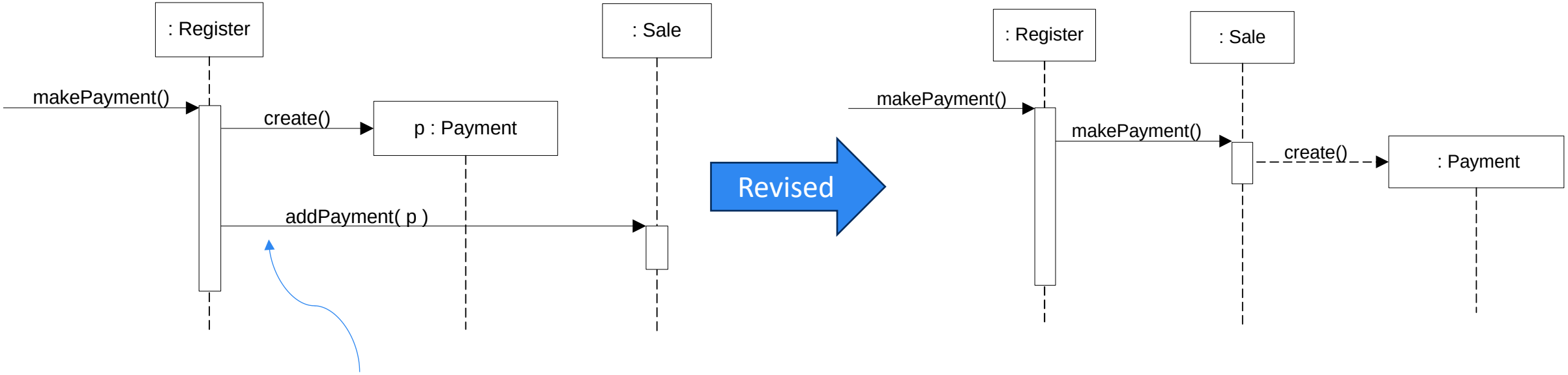


Another Example: POS

Why Sale is responsible for creating Payment? Why not Register?

In the real-world domain, Register *records* the Payment. By Expert, Register should be responsible for creating Payment.
→ *Yes.* It is acceptable.
→ BUT, in the real-world domain, Register involves most of the related system operations → *Register is potentially bloated*



Since Register has to addPayment to Sale, let's delegate the creation of Payment responsibility to Sale!

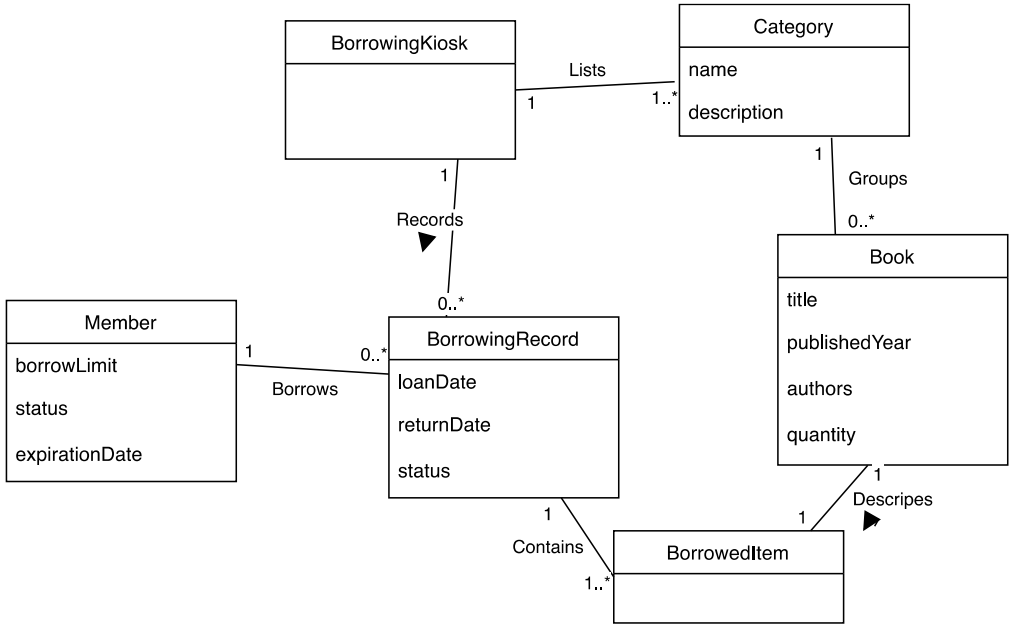
High Cohesion: Discussion

- Similar to Low Coupling, High Cohesion is an ***evaluative principle*** that should be kept in mind during all design decisions
- Both Low Coupling and High Cohesion must be considered together when designing software objects

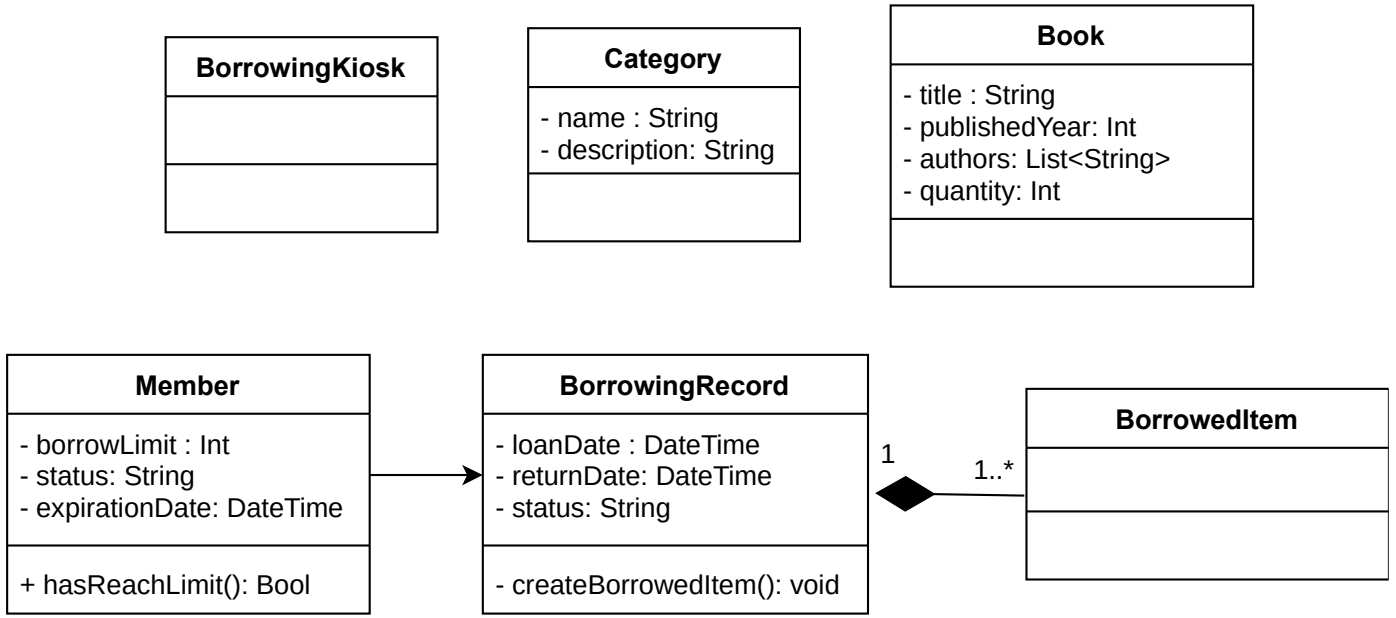
Contraindications

- Low cohesion (=many responsibilities in one class) sometimes is necessary for the design to meet strict non-functional requirements
  - Ex: Larger and less cohesive classes are used to reduce processing overheads (e.g., transmission, storage) to meet performance requirements

Example: Designing the design model based on the domain model



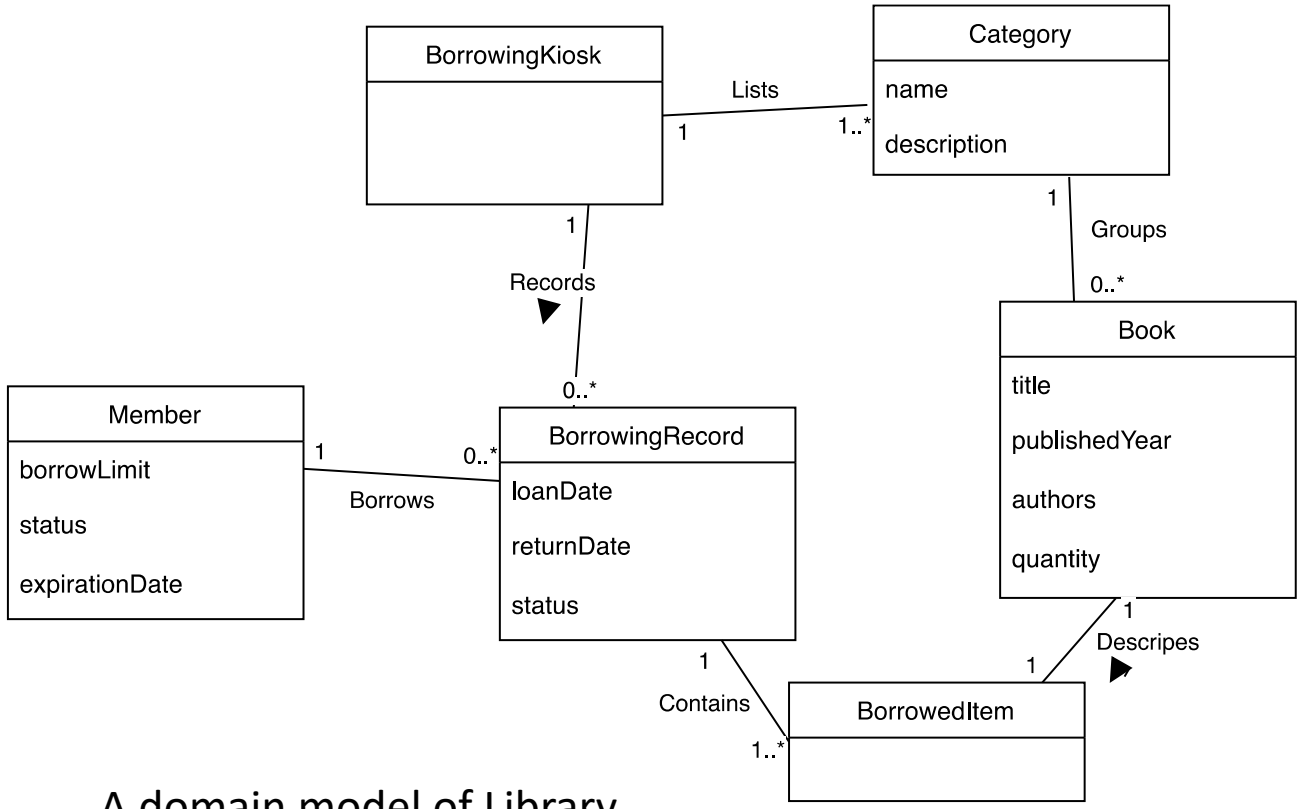
A **domain model** of Library Borrowing System



An *incomplete* **design model** of Library Borrowing System

Why can't the BorrowingKiosk class have responsibility of "hasReachedLimit" instead of the Member class?

Exercise: Assigning responsibilities to software objects of the design model based on the domain model



A domain model of Library Borrowing System

BorrowingKiosk can also access BorrowingRecord.
Should we assign the responsibility of "hasReachedLimit" to the BorrowingKiosk class instead of the Member class?

Discussion:

- Yes. It is acceptable.
- But it can increase coupling (i.e., an association between BorrowingKiosk and Member is needed in the design model)
- Also, BorrowingKiosk is likely to handle many operations -> increase a chance of low cohesion

Summary & Remarks

- GRASP is a set of patterns/principles that aid us to design software objects to be more maintainable
- GRASP guides us in making decision about ‘who should be responsible for doing/knowing something
- Given the same domain model/problem, there can be many possible design solutions that meets GRASP principles
- When there are many possible design solutions

  → Use Low Coupling and High Cohesion to evaluate the design solutions

Lecture Identification

Lecturer: Philip Dart

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These slides include materials from:

Applying UML and Patterns: An Introduction to Object-Oriented Analysis and Design and Iterative Development, Third Edition, by Craig Larman, Pearson Education Inc., 2005.